UCLA ANDERSON MSBA VIBATHON 2026 · THEME 6 · LANDED COST

SourceSwitch AI

Knowing when a tariff change should change your supplier.

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A landed-cost decision tool for small and mid-sized U.S. importers

THE PROBLEM, AND WHAT WE ARE BUILDING

Two years of tariff whiplash. Small importers have the least power to push back.

Since early 2025 U.S. tariffs have moved from +10% to +145% and back, and about 94% of the cost landed on U.S. importers. Small and mid-sized businesses have little bargaining power with suppliers and little room to raise prices. Every change forces the same question: **is it time to switch supplier country?**

SourceSwitch AI takes

Current and alternative origin, supplier unit quotes, freight per unit, tariff rates by origin, order quantity.

and returns

Landed cost per unit side by side, the break-even tariff at which the ranking flips, and potential savings.

one clear flag

Stay · Switch · Renegotiate

DATASET

Two sources of real U.S. import data, by product and country of origin

Consumer goods · UN Comtrade

Grain: HS6 × origin country, 2023–2025 plus Jan–Feb 2026. U.S. as reporter; same Census data as USITC DataWeb.

Fields: CIF value, FOB value, reported quantity, import share.

Scope: 36 categories across furniture, apparel, footwear, bags, toys, kitchenware and electronics.

Industrial goods · USITC DataWeb

Grain: HTS10 × origin country, 2023–2025. 49,913 observations from USITC DataWeb.

Fields: customs value, CIF, quantity and **actual duties collected**, giving each origin's real 2025 effective tariff.

Scope: 13 curated products across automotive parts, electronics and household products.

Unit CIF

= CIF value ÷ quantity, the market-average cost of one unit at a U.S. port

Landed

= customs value × (1 + tariff) + freight and insurance

Only lines with a reported physical quantity are usable. Weight-only lines (toys, bags, kitchenware, lamps, bedding) were dropped: Comtrade back-fills their quantity with one uniform unit value, so every origin looks identical.

Unit values are market averages, not supplier quotes. We use them to find the market, and the importer's own quotes to make the decision.

STEP 1 · NARROWING THE TARGET

Same screen on both datasets: keep categories where origin costs are close

Consumer goods · Comtrade HS6, 2024

36	23	7	4
pulled	comparable	near-tied	multi-country

Industrial goods · USITC HTS10, 2025

49,913	13	7	5
observations	curated	near-tied	multi-country

Comparable: a physical quantity is reported, so unit cost can be compared across origins.

Near-tied: the two cheapest origins with ≥3% import share are within 20% on unit cost.

Multi-country: three or more origins sit within 20% of the cheapest, so any tariff, currency or quote move can reorder them.

Why closeness? The break-even tariff that flips a sourcing decision equals the cost gap. A 4% gap flips on a 4-point tariff change; a 300% gap never does. Result: **9 target categories across 5 industries**.

OUR TARGET · 9 CATEGORIES IN 5 INDUSTRIES

Where three or more countries compete within 20% on cost, so a tariff move flips the answer

Women’s knit cotton trousers

Apparel · HS 610462 · 5 origins ≤20%

Nicaragua $2.8 vs China $2.9 · **1% gap**

Leather footwear

Footwear · HS 640399 · 4 origins ≤20%

China $16.5 vs Indonesia $17.1 · **4% gap**

Cotton T-shirts

Apparel · HS 610910 · 4 origins ≤20%

Dom. Rep. $1.4 vs Haiti $1.4 · **4% gap**

Wooden furniture

Furniture · HS 940360 · 3 origins ≤20%

Canada $15.7 vs Italy $17.5 · **11% gap**

Plastic brake hoses

Automotive parts · HTS 8708.99.81 · 3 origins ≤20%

Spain $9.1 vs Mexico $9.4 · **3% gap** · China paid 50%, Mexico 9%

CNC machine-tool controls

Electronics · HTS 8537.10.91 · 6 origins ≤20%

China $3,476 vs Japan $3,602 · **4% gap** · China paid 38%, Japan 11%

Heater parts

Electronics · HTS 8516.90.05 · 3 origins ≤20%

China $2.65 vs Mexico $2.86 · **8% gap** · China paid 51%, Mexico 0%

Bumper parts

Automotive parts · HTS 8708.10.60 · 4 origins ≤20%

China $11.9 vs Japan $13.3 · **12% gap** · China paid 60%, Japan 18%

Cast-iron suspension parts

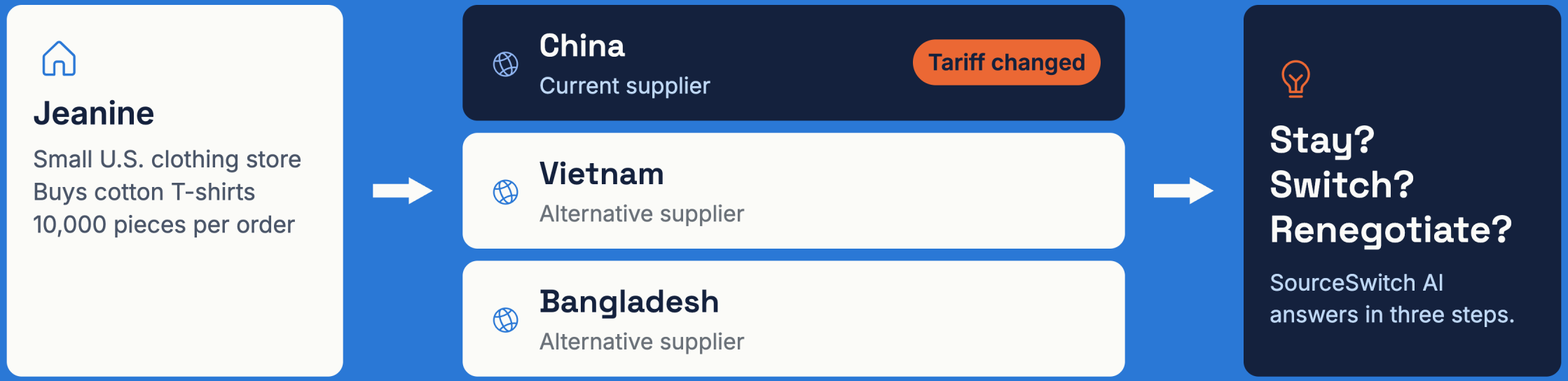
Automotive parts · HTS 8708.80.60 · 3 origins ≤20%

Brazil $2.79 vs S. Korea $3.13 · **12% gap** · China paid 56%, Brazil 6%

Blue: UN Comtrade 2024 unit CIF with the 2026 tariff table. Orange: USITC DataWeb 2025 unit CIF and actual duties paid. Origins with ≥3% share; gap = next-cheapest vs cheapest.

THE PRODUCT · LIVE DEMO

Jeanine runs a clothing store. China's tariff just changed. What should she do?



Appendix

APPENDIX · WHY WE NARROWED

For many products one country has an absolute cost advantage, and no tariff changes the answer

Headphones · HS 851830

381%

China $10.4 vs Vietnam $50.0 per unit. China stays cheapest at any tariff seen so far.

Metal-frame upholstered seats · HS 940171

369%

China $20.8 vs Vietnam $97.4 per unit.

Bicycles · HS 871200

325%

China vs Cambodia, the next major origin.

So we used real U.S. import data to find the categories where origin costs are **close enough that a tariff, currency or freight move can flip the ranking**. Those importers are our target customers.

Gap = unit CIF of the next-cheapest major origin vs the cheapest, U.S. general imports 2024 (UN Comtrade)

APPENDIX · DOES CLOSENESS MATTER?

In the four, a 10-point tariff or a 10% currency move flips the cheapest origin. In wide-gap categories nothing does.

Category	Cost gap	+10 pt tariff on cheapest	+10% currency on cheapest	Ocean freight +86%
Women's knit cotton trousers	1%	Flips	Flips	No change
Leather footwear	4%	Flips	Flips	No change
Cotton T-shirts	4%	Flips	Flips	No change
Wooden furniture, other	11%	Flips at +11 pt	Flips at +11%	No change
16 wide-gap categories (23–381%)	23–381%	No change	No change	No change

Freight is not the lever. Freight + insurance is 3–12% of FOB for these goods, so even the 86% jump in the Freightos index this year moves landed cost by only a few percent.

And it happened. After the 2025 tariffs China lost 30% of its 2024 share in cotton T-shirts and leather footwear, and 18% in wooden furniture. Importers in near-tied markets moved.

Shocks applied to 2024 unit CIF of the cheapest major origin; freight shock from FBX global index $1,850 → $3,440/FEU (Feb–Nov 2026). Share change: U.S. import value 2024 → 2025.

APPENDIX · DATASET LIMITATIONS

What the data can and cannot tell us

Unit values are averages

- An HS6 or HTS10 line mixes product grades, so a "close" unit cost can hide a quality gap (China $10 earbuds vs Vietnam $50).
- They are market benchmarks, not supplier quotes. The tool decides on the importer's own quotes.
- Only lines with a reported physical quantity are usable; weight-only lines were dropped.

Tariffs are two different things

- Consumer categories: a 2026 tariff table reconstructed from CBP and USTR notices. Not legally verified; AD/CVD orders and Section 301 exclusions are not modelled.
- Industrial categories: the 2025 effective duty actually paid (USITC duties ÷ customs value). Real, but historical, and it blends regimes within the year.
- USMCA qualification is assumed for Mexico and Canada.

Coverage and sample size

- 2026 trade data is Jan–Feb only; ten consumer categories have 2023–2024 only (Comtrade's free quota).
- The 13 industrial products were hand-curated, not screened from the whole tariff schedule; the 36 consumer lines are a judgment sample too.
- Origins below 3% import share are excluded, so a small cheap origin can be missed.

What we would do next

Pull the full 2026 series and the remaining HS lines · screen the whole tariff schedule, not a curated list · verify current rates line by line against the HTSUS and add AD/CVD and exclusions · let importers upload quotes for several origins and track their thresholds over time.